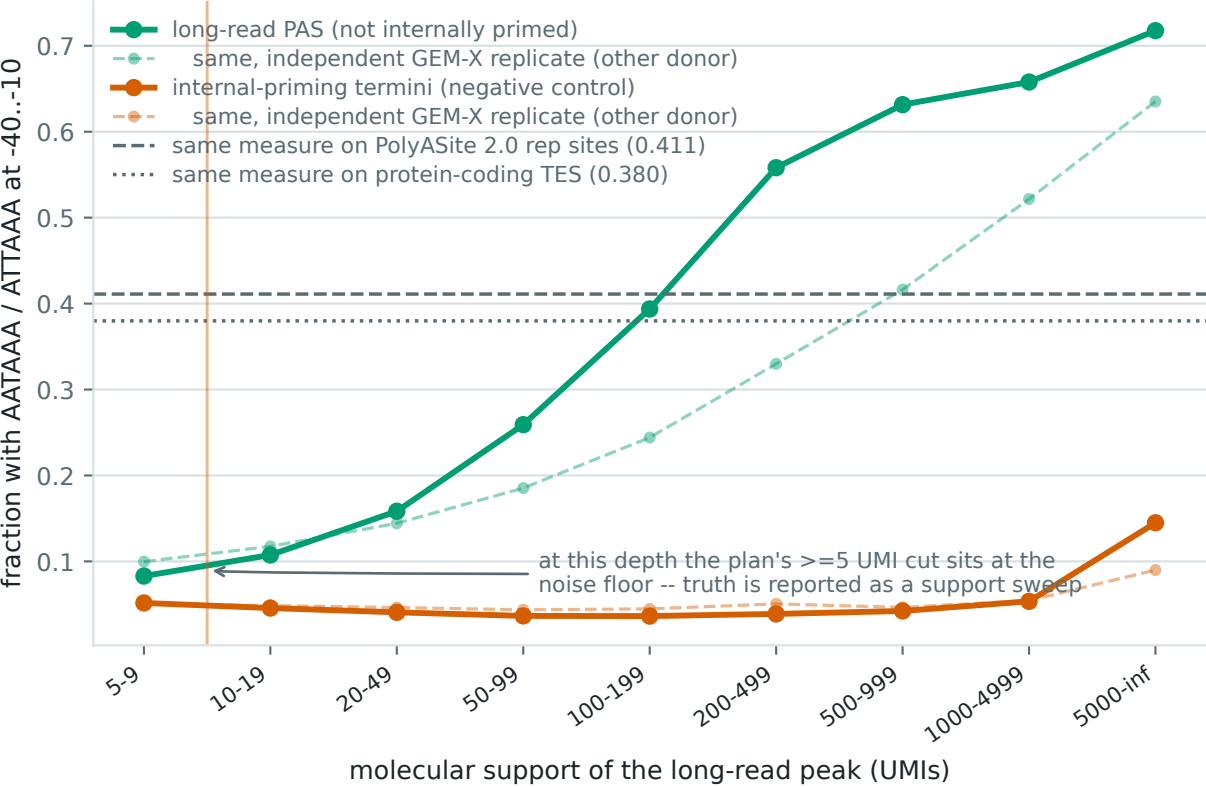
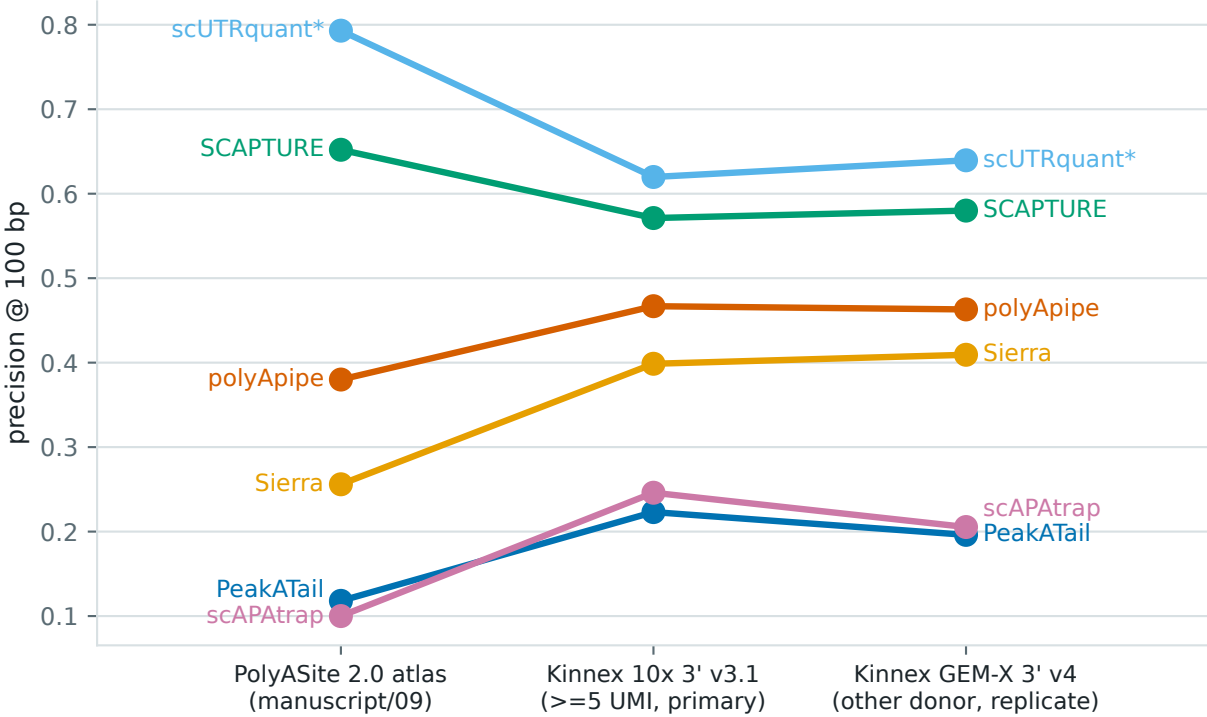


Atlas-independent validation of the pbmc_10k_v3 head-to-head against PacBio Kinnex single-cell long reads

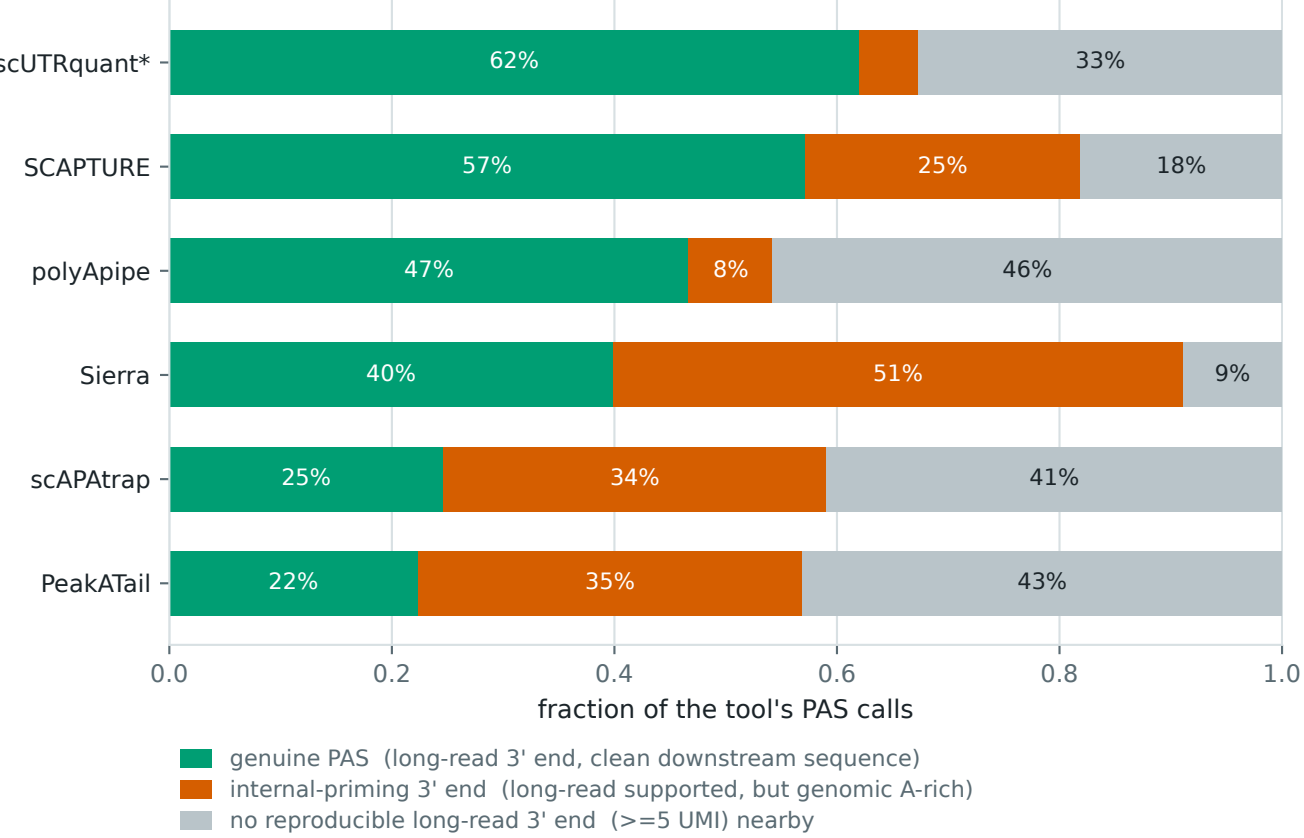
A The truth set is real, but its support threshold is depth-dependent



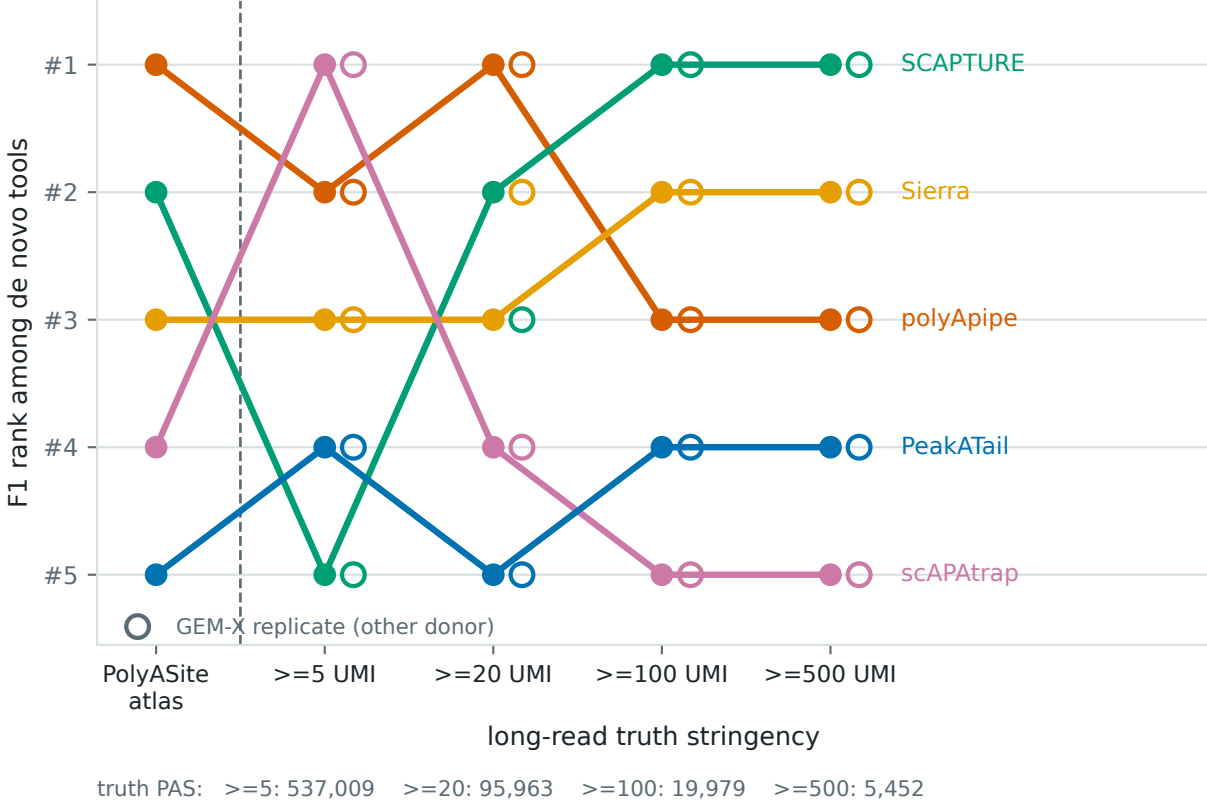
B Precision ordering replicates (rho = 0.90 vs atlas, 1.00 between donors)



C What each tool's calls actually sit on (100 bp, strand-matched)



D Atlas F1 ranking does not survive; long-read ranking reproduces



expression-matched control -- restricted to calls inside genes carrying a >=100-UMI long-read PAS, 'no long-read 3' end' stays
PeakATail 40%, polyApipe 38%, Sierra 6%, SCAPTURE 3%: not a coverage artifact
independent GEM-X replicate (other donor, 3' v4), genuine / internal-priming / none: PeakATail 20/30/51%; Sierra 41/50/9%; SCAPTURE 58/22/20%

CAVEAT Neither Kinnex donor is the pbmc_10k_v3 donor and the 10x chemistries differ (Kinnex 10x 3' v3.1 and GEM-X 3' v4 vs pbmc_10k_v3 3' v3): this is SITE-LEVEL truth. It validates PAS POSITIONS, not per-cell usage. Primary truth = 76.8 M dedup FLNC molecules from 12,852 cells: isoseq refine --require-polya, minimap2 -ax splice:hq -uf, real-cell CB, 3' softclip <=30 nt, greedy +/-25 nt peak calling on the terminus histogram. Internal priming = >=12 A or >=6 consecutive A in the 18 genomic nt downstream of the cleavage site, read in transcript direction. No threshold anywhere was chosen using an atlas; atlas agreement is only reported. F1 compares a fixed call set against truth sets differing 99-fold in size, so its ranking moves with stringency even though it reproduces between donors at a given stringency; the precision ordering (B) does not. scUTRquant* is annotation-based (a fixed UTRome catalog filtered by detection): it is shown for scale but is not ranked with the de novo tools.